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A fast recovery pulsed nuclear magnetic resonance sample probe using a delay line*

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A sample probe for pulsed nuclear magnetic resonance that replaces the normal resonant circuit by a lumped parameter delay line is described. Theoretical analysis shows that the delay line probe has a signal-to-noise ratio and conversion efficiency of power into rotating magnetic field equivalent to a resonant circuit of $Q = (2\pi)$ (delay time) (Larmor frequency). However, the delay line probe has a much wider bandwidth, much shorter transient decay time, and faster recovery from the rf excitation pulse. Experimental results are included that demonstrate submicrosecond recovery times for the observation of the signal.

I. INTRODUCTION

In this paper we describe a sample probe for pulsed nuclear magnetic resonance (PNMR) measurements that does not use any resonant circuits. The basic component of this probe is a lumped parameter delay line having a frequency response that is essentially flat from 0 to its cut-off frequency. This type of unit has the advantage of the high signal-to-noise ratio of probes using resonant circuits, without their corresponding disadvantages of tuning, narrow signal bandwidth, and poor transient response to the excitation pulse (that is long "ringing" times).

For pnmr spectroscopy of solids, a variety of passive and dynamic damping devices for sample probes using resonant circuits have been described in the literature in recent years.¹⁻¹² Some of the functions of the probe and associated circuitry are:

- (a) The efficient conversion of the rf power from the transmitter into an intense, homogeneous radio frequency (rf) magnetic field in the space occupied by the sample.
- (b) The rapid build up and decay of the rf magnetic field, both for delineating an origin for the free-induction-decay (fid) produced by the sample, and for observing it before it dies out.
- (c) Isolation of the receiver amplifier from the rf excitation pulse to protect it from the large rf voltages and to facilitate its fast recovery from the pulse.
- (d) The production of a strong fid signal from the sample and efficiently coupling it to the receiver.

A parallel resonant circuit consisting of an inductance L_0 , a capacitance C_0 and a parallel resistance R_0 has a "ringing" transient of angular frequency ω_r , and decay time constant τ_r of

$$\omega_r = \omega_0 [1 - 1/(4Q^2)]^{1/2} \quad (1)$$

$$\tau_r = 2Q/\omega_0, \quad (2)$$

where $\omega_0 = [L_0 C_0]^{-1/2}$ and $Q = R_0/(\omega L_0)$. Equation (2) implies that in order to satisfy requirement (b) in a single coil probe, the circuit Q should be kept as low as possible and the resonant frequency as high as possible. The Q cannot be made arbitrarily small however, because the voltage signal-to-noise ratio¹³ obtained from the tuned circuit is proportional to $Q^{1/2}$. During the application of the rf excitation pulse, the value of the linearly polarized rf magnetic field B_1 inside the coil in mks units (neglecting fringing fields and filling factors) is given by²

$$B_1 = [2\mu_0 P Q / (V \omega_0)]^{1/2}, \quad (3)$$

where P is the output power from the transmitter into a resistive load R , V is the volume of the coil, and $\mu_0 = 4\pi \times 10^{-7}$ H/m. Equation (3) indicates that again Q cannot be made arbitrarily small if condition (a) is to be satisfied.

A "crossed-coil" probe does not resolve, in practice, the problems arising from the conflicting requirements (b) and (a) and (d).¹² In a tuned single coil probe, the rf excitation voltage has to decrease from about 10^3 V to about 10^{-5} V before the nuclear signal can be observed without distortion. Since the broadband isolation that is achievable with a crossed coil probe is typically only of the order of 60 dB, little improvement in the over-all recovery time is obtained. To avoid the ringing of the tuned receiver coil for long times, one would still have to keep the Q of that coil relatively low.

We have attempted to satisfy requirements (a), (b), and (d) by designing a probe for pnmr that uses a lumped parameter delay line in place of a tuned circuit. For this type of probe, it will be shown in Sec. II that there is no conflict of requirement (b) with (a) and (d). Since no tuning capacitor is required close to the sample coil, the probe is mechanically simpler. This advantage is traded for the greater effort needed to construct the delay line coil. Because of the wide bandwidth of the delay line probe, it

should be useful for experiments where the sample is irradiated with several different frequencies.

In Sec. II A–C, we develop the theory for the response of the lumped parameter delay line to the excitation pulse from the transmitter, and to the subsequent signals produced by the sample. In Section IID we calculate the Q of a resonant circuit that is equivalent to our delay line in producing an oscillating magnetic field and picking up a signal from a sample.

In Sec. III, we describe the particular electrical circuit configuration used to couple the delay line to the transmitter and the receiver; in Sec. IV, we discuss the design parameters of the delay line we have used for the past two years. For this geometry, the rf magnetic field is relatively inhomogeneous. For some experiments such as line shape studies, spin lattice relaxation time measurements where several saturating pulses are used, and several types of rotating frame measurements, an inhomogeneous rf magnetic field can be tolerated. For those experiments where a homogeneous rf magnetic field is needed, a more complicated delay line geometry can be used as will be indicated in Sec. IV, where the performance characteristics of a delay line probe are presented.

II. THEORY OF THE LUMPED PARAMETER DELAY LINE

For the analysis of the lumped parameter delay line, it will be assumed that the unit consists of N identical π sections arranged as a low pass filter, as shown in Fig. 1. L_1 is the inductor that represents one section of the sample coil, there being N sections. To keep the mathematical analysis simple, it will be assumed that there is no magnetic coupling between the various inductive sections. This assumption is not exactly true for many delay line configurations, but it should not significantly affect the conclusions drawn. C_1 represents one-half the value of the lumped capacitor between coil and ground used between each coil section. The voltage generator ϵ_n represents the signal picked up by the n th section of the delay line from the excited sample after the excitation pulse is turned off. It will be assumed equal to zero for the analysis while the excitation pulse is on. Only the steady state response of the delay line will be examined. The transient response of the delay line to a rf pulse can be found by taking the Fourier transform of the product of Fourier transform of the rf pulse and steady state response function of the delay line. This is a complicated calculation and not necessary to this paper, but a few remarks will be made about the subject at the end of Sec. IIA.

A. Transfer Matrix

The analysis of the steady state solutions of lumped parameter delay lines for the case where $\epsilon_n = 0$ [Fig. 1(b)], can be found in a number of places.¹⁴ For an arbitrary ϵ_n the nodal equations for the delay line section of Fig. 1(b) are

$$I_{n-1} - Y_1 E_{n-1} - Z_1^{-1}(E_{n-1} + \epsilon_n - E_n) = 0 \quad (4)$$

$$Z_1^{-1}(E_{n-1} + \epsilon_n - E_n) - Y_1 E_n - I_n = 0, \quad (5)$$

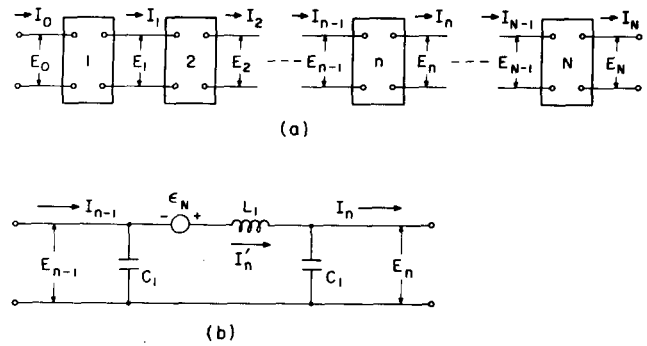


FIG. 1. Simplified model for the analysis of the delay line sample probe. (a) Arrangement of the individual π sections to form the delay line and notation used in the analysis. (b) The n th π section, consisting of an inductance L_1 that represents one section of the sample coil and two capacitors C_1 , each being one-half of the lumped capacitance to ground placed between each coil section. The voltage generator ϵ_n represents the excitation signal induced in the n th coil section by the sample. It will be set equal to zero for the analysis of the delay line during the excitation pulse.

where

$$Z_1 = i\omega L_1 \quad (6)$$

$$Y_1 = i\omega C_1$$

$$f = \omega/2\pi = \text{excitation frequency.}$$

Rearranging terms yields

$$\begin{bmatrix} E_n \\ I_n \end{bmatrix} = \epsilon_n \begin{bmatrix} 1 \\ -Y_1 \end{bmatrix} + [M^{-1}] \begin{bmatrix} E_{n-1} \\ I_{n-1} \end{bmatrix} \quad (7)$$

$$[M^{-1}] = \begin{bmatrix} 1 + Y_1 Z_1 & -Z_1 \\ -(2Y_1 + Y_1^2 Z_1) & 1 + Y_1 Z_1 \end{bmatrix}. \quad (8)$$

The symbols a (propagation parameter), Z_0 (characteristic impedance), and ω_c (cutoff angular frequency) are defined as

$$\omega_c = [2/(L_1 C_1)]^{1/2} \quad (9)$$

$$\begin{aligned} \cosh a &= 1 + Y_1 Z_1 \\ &= 1 - 2\omega^2/\omega_c^2 \end{aligned} \quad (10)$$

$$\begin{aligned} Z_0 &= \left(\frac{Z_1}{Y_1(2 + Y_1 Z_1)} \right)^{1/2} \\ &= \left\{ \left(\frac{L_1}{2C_1} \right) \left[1 - \left(\frac{\omega}{\omega_c} \right)^2 \right]^{-1} \right\}^{1/2}. \end{aligned} \quad (11)$$

We may then rewrite $[M^{-1}]$ as

$$[M^{-1}] = \begin{bmatrix} \cosh a & -Z_0 \sinh a \\ -Z_0^{-1} \sinh a & \cosh a \end{bmatrix}. \quad (12)$$

It is easily shown that

$$[M^n] = \begin{bmatrix} \cosh na & Z_0 \sinh na \\ Z_0^{-1} \sinh na & \cosh na \end{bmatrix}. \quad (13)$$

B. Response of the Lumped Parameter Delay Line to the Excitation Pulse

During the excitation pulse, the driving voltage is much larger than all the ϵ_n 's and they will be ignored for this part

of the calculation. One concludes from Eq. (7) that

$$\begin{bmatrix} E_n \\ I_n \end{bmatrix} = [M^{-1}] \begin{bmatrix} E_{n-1} \\ I_{n-1} \end{bmatrix} = [M^{-2}] \begin{bmatrix} E_{n-2} \\ I_{n-2} \end{bmatrix} \quad (14)$$

$$\begin{bmatrix} E_n \\ I_n \end{bmatrix} = [M^{-n}] \begin{bmatrix} E_0 \\ I_0 \end{bmatrix}, \quad (15)$$

and by the same arguments

$$\begin{bmatrix} E_n \\ I_n \end{bmatrix} = [M^{N-n}] \begin{bmatrix} E_N \\ I_N \end{bmatrix}. \quad (16)$$

For the time that the excitation pulse is applied, we shall assume that the delay line is terminated in its characteristic impedance Z_0 so that

$$I_N = Z_0^{-1} E_N. \quad (17)$$

With the application of Eq. (13), Eq. (16) may be reduced to

$$\begin{aligned} \begin{bmatrix} E_n \\ I_n \end{bmatrix} &= E_N \begin{bmatrix} \cosh(N-n)a & Z_0 \sinh(N-n)a \\ Z_0^{-1} \sinh(N-n)a & \cosh(N-n)a \end{bmatrix} \begin{bmatrix} 1 \\ Z_0^{-1} \end{bmatrix} \\ &= E_N e^{(N-n)a} \begin{bmatrix} 1 \\ Z_0^{-1} \end{bmatrix} = E_0 e^{-na} \begin{bmatrix} 1 \\ Z_0^{-1} \end{bmatrix}. \end{aligned} \quad (18)$$

For $\omega < \omega_c$, $a = ib$ is a pure imaginary number and the signal at node n lags behind the signal at the input by the phase factor nb , but is otherwise identical to it. This is true for $n=0$, so that the input impedance of the delay line terminated in Z_0 , is just Z_0 , which is, relatively, frequency independent for $\omega \ll \omega_c$.

From Eq. (10), we may solve for the phase lag b as

$$\begin{aligned} b &= \sin^{-1} \left\{ 2 \left(\frac{\omega}{\omega_c} \right) \left[1 - \left(\frac{\omega}{\omega_c} \right)^2 \right]^{\frac{1}{2}} \right\} \\ &\cong 2 \left(\frac{\omega}{\omega_c} \right) \left[1 + \frac{1}{6} \left(\frac{\omega}{\omega_c} \right)^2 \right] \end{aligned} \quad (19)$$

for $(\omega/\omega_c) \ll 1$. This corresponds to an equivalent time delay per section, denoted by T_1 , of

$$T_1 = \frac{b}{\omega} = (2/\omega_c) \left[1 + \frac{1}{6} \left(\frac{\omega}{\omega_c} \right)^2 \right]. \quad (20)$$

For $\omega \ll \omega_c$, T_1 is independent of ω and the delay line appears nondispersive. With this limitation, rf excitation pulses which have a broad frequency spectrum will travel down the transmission line without much distortion. The only place one needs a more detailed analysis is the behavior of the trailing edge of the rf pulse. As mentioned earlier, transient analysis of a lumped parameter delay line is extremely complicated. However, analysis¹⁵ of the response of single sections of the delay line, terminated in their characteristic impedance, to a step voltage shows that they ring at a frequency near the cutoff frequency $(\omega_c/2\pi)$ with a decay time of the order of $(2/\omega_c)$. One expects lower frequency components to appear for a multisection delay line, but we experimentally find these to last only several

times $(2/\omega_c)$ and have not found them to be an experimental problem.

C. Response of the Lumped Parameter Delay Line to the Sample Signal

To compute the signal voltage delivered to a receiver from the sample, we again start with Eq. (7). By successively using Eq. (7) for $n=1, 2, 3, 4 \dots N$ one finds

$$\begin{bmatrix} E_N \\ I_N \end{bmatrix} = \left(\sum_{n=1}^N \epsilon_n [M^{N-n}] \begin{bmatrix} 1 \\ -Y_1 \end{bmatrix} \right) + [M^{-N}] \begin{bmatrix} E_0 \\ I_0 \end{bmatrix}. \quad (21)$$

Let us now assume that the end of the delay, line connected to the transmitter, is terminated in a source impedance Z_s so that $I_0 = -E_0/Z_s$, and, the receiver end of the delay line is terminated in an impedance Z_R so that $I_N = E_N/Z_R$. Eliminating I_0 , I_N , and E_0 from Eq. (21) yields the general solution for E_N listed below,

$$E_N = D \left[(1 - Y_1 Z_0) E_+ + (1 + Y_1 Z_0) \left(\frac{Z_0 - Z_s}{Z_0 + Z_s} \right) E_- \right], \quad (22)$$

where

$$D^{-1} = \left(1 + \frac{Z_0}{Z_R} \right) e^{Na} + \left(1 - \frac{Z_0}{Z_R} \right) \left(\frac{Z_0 - Z_s}{Z_0 + Z_s} \right) e^{-Na}, \quad (23)$$

$$1 - Y_1 Z_0 = 2(e^a + 1)^{-1}, \quad (24)$$

$$1 + Y_1 Z_0 = 2(e^{-a} + 1)^{-1}, \quad (25)$$

$$E_+ = \sum_{n=1}^N \epsilon_n e^{na}, \quad (26)$$

$$E_- = \sum_{n=1}^N \epsilon_n e^{-na}. \quad (27)$$

E_+ is the excitation wave, produced by the interaction of the excited sample with the delay line, traveling from the transmitter end of the delay line to the receiver end and E_- is the excitation wave traveling in the other direction. In the previous section, it was shown that when the receiver end of the delay line was terminated with impedance Z_0 and $\omega \ll \omega_c$, the n th section of the delay line is excited by an rf pulse identical to the rf pulse at the transmitter end of the delay line, except delayed by a phase nb . The response of the n th section should also be delayed by the same phase factor, and assuming that each section of the delay line contains the same amount of sample and behaves in the same way, we may write

$$\epsilon_n = \epsilon_0 e^{-na}, \quad (28)$$

where ϵ_0 is the amplitude of the signal produced by any one section of the line. Thus, with this assumption

$$E_+ = N \epsilon_0, \quad (29a)$$

$$E_- = \epsilon_0 \sum_{n=1}^N (e^{-2ib})^n = \epsilon_0 (e^{-ib(N+1)}) \frac{\sin Nb}{\sin b}. \quad (29b)$$

For a delay line where $Nb \gg \pi$, $E_- \ll E_+$ and E_- can usually be ignored. This is done for the evaluation of E_N in Eq. (22)

for some of the special cases listed below:

$$Z_s = Z_0, \quad Z_R = Z_0 \quad E_N = e^{-Na}(e^a + 1)^{-1}E_+, \quad (30)$$

$$Z_s = Z_0, \quad Z_R = \infty, \quad E_N = 2e^{-Na}(e^a + 1)^{-1}E_+, \quad (31)$$

$$Z_s = \infty, \quad Z_R = Z_0, \quad E_- = 0, \quad E_N = e^{-Na}(e^a + 1)^{-1}E_+, \quad (32)$$

$$Z_s = 0, \quad Z_R = Z_0, \quad E_- = 0, \quad E_N = e^{-Na}(e^a + 1)^{-1}E_+. \quad (33)$$

Equations (30) and (31) imply that for $Z_s = Z_0$, E_N is independent of E_- and insensitive to Z_R as long as $Z_R > Z_0$. Equations (32) and (33) imply that for $Nb \geq \pi$ and $Z_R = Z_0$, E_N is independent of Z_s . Both of these conclusions are reasonable.

D. Comparison of a Lumped Parameter Delay Line and a Resonant Circuit

Pulsed NMR apparatuses usually use resonant circuits to produce oscillating magnetic fields in the space occupied by the sample, and to pick up signals generated by the sample. It is, therefore, of interest to compare the efficiency with which a lumped parameter delay line does these two things with that of a resonant circuit. For this comparison, the N inductances of our transmission line (each having inductance L_1 , each filled with the same sample material, and each having zero magnetic coupling to the other $N-1$ inductances) are reconnected as a single inductor of value $L_N = NL_1$. This is resonated by a capacitor C_N at the angular frequency ω and has in series with it a resistor R_N , as shown in Fig. 2, so that the Q of the circuit, denoted by Q_N is

$$Q_N = \omega L_N / R_N. \quad (34)$$

First we shall compute the value for Q_N so that the same amount of power P_R delivered to the resonant circuit produces the same magnitude of oscillating magnetic field. Since the magnetic coupling between inductors is assumed to be zero, the magnitude of the oscillating magnetic field inside each of the inductors in the two configurations is the same if the currents flowing through them are the same. For the resonant circuit, the power P_R applied to the resonant circuit and dissipated in R_N is

$$P_R = \frac{1}{2} I_R^2 R_N = \frac{1}{2} I_R^2 \left(\frac{\omega L_N}{Q_N} \right), \quad (35)$$

$$P_R / I_R^2 = \frac{1}{2} \omega N L_1 / Q_N. \quad (36)$$

For the transmission line, we see that from Fig. 1(b), the current I_n' flowing through the n th inductance of the transmission line is

$$I_n' = (E_{n-1} - E_n) / Z_1 \\ = E_0 e^{-nib} (e^{ib} - 1) / Z_1. \quad (37)$$

Thus

$$I_n' (I_n')^* = 2E_0 E_0^* (1 - \cos b) / (\omega L_1)^2. \quad (38)$$

The power P_D applied to the delay line terminated in Z_0 is

$$P_D = E_0 E_0^* / (2Z_0). \quad (39)$$

Using the relation $Z_0 \sin b = \omega L_1$, we find

$$\frac{P_D}{I_n' (I_n')^*} = \frac{\omega L_1 \sin b}{4(1 - \cos b)}; \quad (40)$$

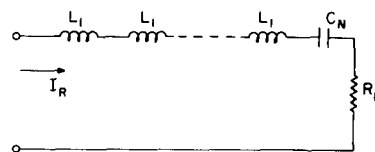


FIG. 2. N inductors of the delay line reconnected as a resonant circuit for generating an oscillating magnetic field. The magnetic coupling between the inductors is assumed to be zero, thus, $L_N = NL_1$.

setting

$$\frac{P_D}{(I_n') (I_n')^*} = \frac{P_R}{I_R^2} \quad (41)$$

yields

$$Q_N = 2N \left(\frac{1 - \cos b}{\sin b} \right) \\ = 2N \left(\frac{\omega}{\omega_c} \right) \left[1 - \left(\frac{\omega}{\omega_c} \right)^2 \right]^{-\frac{1}{2}}. \quad (42)$$

For $\omega \ll \omega_c$, the above expression reduces to

$$Q_N = \omega N T_1 = \omega T_N \quad (43)$$

where T_N is the total time delay of the delay line. Equation (43) states that the equivalent Q of a delay line for the production of an oscillating magnetic field is just 2π times the number of rf cycles stored in the delay line. A way of viewing this result is that the efficiency of storing energy in a resonant circuit for the production of large oscillating magnetic fields is equivalent to the increase in energy density of a traveling wave on a delay line due to the slowing down of its speed and the subsequent shortening of its wavelength.

The computation of Q_N for a resonant circuit picking up a signal from the sample is carried out in a manner similar to the previous calculation, and the equivalent circuit for this is shown in Fig. 3. We again assume no magnetic coupling between the N sections of the inductance L_N , so the voltage ϵ_n induced in the n th coil section of the inductance is independent of the behavior of the other sections of the sample, and the voltage generated in L_N by the oscillating magnetic field of the sample is just $N\epsilon_0$. As shown in Fig. 3, the resonant circuit with source resistance R_N is connected to the receiver having input resistance R_N through a coaxial line of impedance R_N . The purpose of the transmission line connecting the resonant circuit to the receiver is to put the resonant circuit on equal footing with the lumped parameter delay line which is terminated in its characteristic impedance and which allows the pick up circuit to be far away from the receiver, coupled to it by a coaxial line. The load is chosen equal to R_N to allow for maximum power transfer to it from the source, of resistance R_N ,

$$I_R = \frac{N\epsilon_0}{2R_N}. \quad (44)$$

Denoting by P_R the power dissipated in the load by the signal induced in the resonant circuit,

$$P_R = \frac{1}{2} I_R^2 R_N = N^2 \epsilon_0^2 / (8R_N), \quad (45)$$

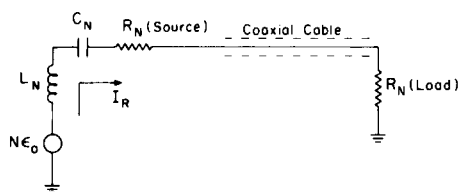


FIG. 3. N inductors of the delay line reconnected as a resonant circuit to produce a signal from the oscillating magnetization of the sample. The magnetic coupling between the inductors is assumed to be zero. R_N represents the source resistance of the resonant circuit. The input impedance of the receiver and the coaxial cable is also assumed to be R_N .

$$\frac{P_R}{\epsilon_0^2} = \frac{NQ}{8\omega L_1} \quad (46)$$

For the case of the lumped parameter delay line terminated in its characteristic resistance, we have from Eqs. (29a) and (30)

$$E_N = N\epsilon_0 e^{-Njb} (1 + e^{jb})^{-1} \quad (47)$$

The power delivered by the sample to the load Z_0 is

$$P_D = \frac{E_N E_N^*}{2Z_0} = \frac{N^2 \epsilon_0^2}{4Z_0 (1 + \cos b)} \quad (48)$$

$$\frac{P_D}{\epsilon_0^2} = \left(\frac{N^2}{4\omega L_1} \right) \left(\frac{\sin b}{1 + \cos b} \right) \quad (49)$$

Since we wish to find the Q_N such that the resonant circuit will deliver the same amount of power as the delay line, we equate P_D and P_R to get

$$Q_N = 2N \left(\frac{\omega}{\omega_c} \right) \left[1 - \left(\frac{\omega}{\omega_c} \right)^2 \right]^{-\frac{1}{2}} \quad (50)$$

the same answer as in Eq. (42) for the production of an oscillating magnetic field with a given amount of power.

For the value of Q_N given above, the power delivered by both these circuits to the receiver from the sample is the same. The signal to noise ratios will also be equal, provided the noise bandwidth for the apparatus using resonant circuit or delay line pick-up coils is limited by narrow banding in the various sections of the receiver system following the probe.

III. SAMPLE PROBE CIRCUIT

A block diagram of our sample probe circuit using a lumped parameter delay line, and the auxiliary circuits that connect it to the transmitter and receiver, is shown in Fig. 4. Our spectrometer operates at 32 MHz and all the coaxial lines and the lumped parameter delay line have characteristic impedances of 50 Ω . The transmitter and receiver have 50 Ω output and input impedances, respectively, so that all impedances are matched.

The transmitter is a gated, untuned, broadband rf power amplifier¹⁶ that delivers 900 W into a 50 Ω resistive load. The output transformer is a North Hills 0500 AA with its unbalanced primary connected to the load. Due to slight differences in the characteristics of the two halves of the 3E29 tube which are connected in push-pull, the input to the balanced secondary of the transformer in the last stage consists of an rf voltage superimposed upon an incompletely cancelled dc pulse. Since some of the high frequency Fourier components of this pulse are within the bandwidth of the transformer but the low frequency components are not, the final output contains a transient of about 15 V lasting about 1 μ sec after the end of a 1 μ sec rf pulse. The quarter wavelength of cable connected to ground at point A in Fig. 4 acts as a short circuit for this low frequency transient and as a high impedance to ground for frequencies close to 32 MHz. The amplitude of the transient is, therefore, greatly reduced without appreciably affecting the rf pulse. It was found that the dc resistance of the cable and the "short" at its end appreciably affected the attenuation

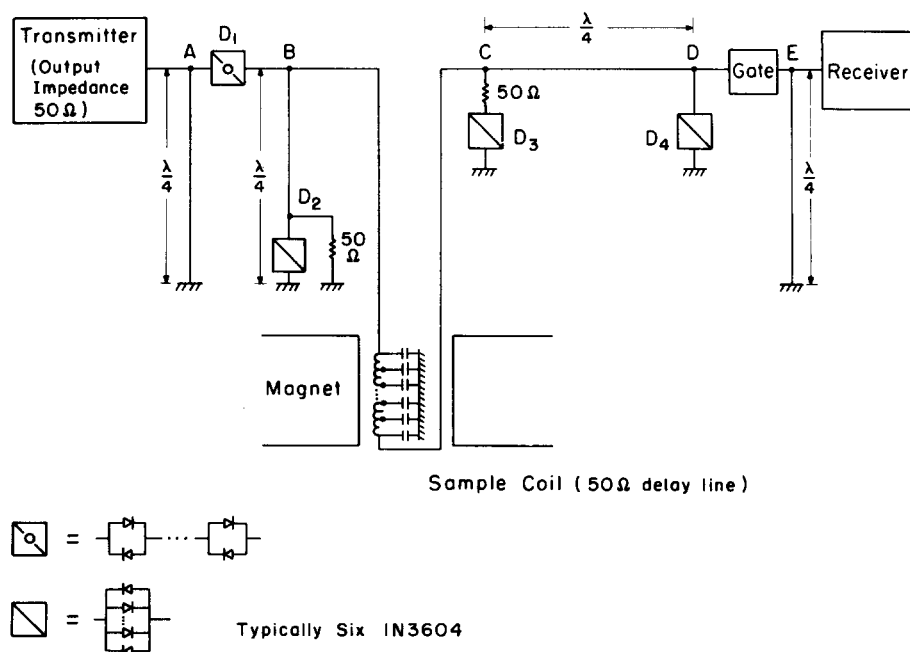


FIG. 4. Block diagram of delay line sample probe circuit.

of low frequencies, thus the $\lambda/4$ cable at point A, and also the one at point E to ground were made of RG-8/U cable.

For the duration of the large amplitude rf excitation pulse, the crossed diodes D_1 , D_2 , D_3 , and D_4 act like low impedances. The quarter wavelength of cable with the crossed diodes D_2 and the $50\ \Omega$ resistor which is connected to point B (Fig. 4), present a high impedance to ground. Similarly at point C, the crossed diodes D_4 and the quarter wavelength of cable also present a high impedance to ground. For high voltage levels, the $50\ \Omega$ transmission line that includes the sample coil is, therefore, terminated in its characteristic impedance.

Following the rf pulse, the crossed diodes D_1 isolate the transmitter from the rest of the system for the ensuing low voltage level signals picked up from the sample. Since the diodes D_2 are now a high impedance, the effective $50\ \Omega$ resistance to ground at point B acts as a matched termination for low voltage level signals traveling in that direction. This prevents them from being reflected back into the delay line by the high impedance of D_1 .

In the forward direction, the nuclear signals generated at different points in the delay line will add in phase as discussed in Sec. IIB. Since diodes D_3 and D_4 also have high impedances, one has in that direction a $50\ \Omega$ transmission line terminated in a matched load of $50\ \Omega$ presented by the receiver input. The line is then completely matched for low voltage levels as assumed in Sec. IIB.

The $\lambda/4$ line between points C and D, along with the diodes D_4 to ground, and the diode gate, all described in detail in Ref. 7, protect the receiver from the rf pulse. The $\lambda/4$ cable from E to ground acts as a filter for any low frequency switching transients generated by the gate. With this arrangement, the over-all isolation of the receiver from the rf pulse at C is about 100 dB.

IV. DESIGN AND CONSTRUCTION OF THE LUMPED PARAMETER DELAY LINE

The lumped parameter delay line was constructed as a single layer close wound solenoid with 10 taps between the ends. AWG 30 copper wire with Formvar insulation was wound on a 0.63 cm diam Teflon form. Two full turns were allowed between taps on the coil, so that the coil had 11 sections of three turns each in a length of 1.0 cm. The coil was then surrounded by a glass tube of 9 mm i.d., containing a longitudinal slit. The space between the glass envelope and the coil was filled with melted sulphur and the Teflon form was removed after solidification. Except for the small amount of Formvar insulation, the sample coil was then free from protons and was therefore satisfactory for studying fluorine and proton resonances at 32 MHz, without any interference from foreign nuclei. We have also constructed similar but more rugged coils by replacing the sulphur with Sauereisen cement¹⁷ (type P 1) mixed with D_2O . The cement seems to absorb water from the air and has to be baked occasionally to get rid of a spurious proton signal.

A compact capacitor bank was constructed by inserting pieces of Kapton H¹⁸ dielectric 0.25 mm thick, between square plates of copper 0.5 mm thick and 20.6 mm wide. Twenty-one plates were used with the top and bottom plates

as well as alternate plates connected to ground. The 10 remaining plates were connected to the taps on the coil. The stack was compressed by a holder consisting of two brass plates joined by two screws. The capacitance could then be adjusted to some extent by the tightness of the screws. The delay line just described actually has coils at the end sections instead of capacitances and should be analyzed as being made up of T sections instead of π sections. The various formulas for a T section delay line are almost the same as for the π section delay line, the only difference is that the Z_0 dependence on f is inverted¹⁴ as shown in Eq. (51).

To make an estimate of the parameters for this delay line using the formulas in Sec. II, we shall ignore the magnetic coupling between coil sections. The measured total inductance of the coil was $3.6\ \mu\text{H}$, and the measured capacity per section was 75 pF. Thus $L_1 = (1/11)(3.6\ \mu\text{H}) = 0.33\ \mu\text{H}$ and $C_1 = \frac{1}{2} \cdot 75\ \text{pF} = 37.5\ \text{pF}$. From Eqs. (9) and (20) and modified Eq. (11),¹⁴ we obtain the parameters

$$f_c = \frac{\omega_c}{2\pi} = 64.0\ \text{MHz},$$

$$Z_0 = 66.4\ \Omega [1 - (f/f_c)^2]^{\frac{1}{2}}, \quad (51)$$

$$Z_0 \text{ (at 32 MHz)} = 57.5\ \Omega,$$

$$T_N = \frac{N}{\pi f_c} [1 + \frac{1}{6} (f/f_c)^2]$$

$$= 54.8 \times 10^{-9} \left[1 + \frac{1}{6} \left(\frac{f}{f_c} \right)^2 \right] \text{ sec}, \quad (52)$$

$$T_N \text{ (at 32 MHz)} = 57 \times 10^{-9} \text{ sec}$$

$$Q_N \text{ (at 32 MHz)} = 12.7.$$

The input impedance of the delay line terminated in a $50\ \Omega$ resistive load was measured using a Boonton¹⁹ R_x meter, type 250A. It was observed to be largely resistive over a wide range of frequencies. The measured input impedance at 32 MHz was a $47\ \Omega$ resistive load in parallel with a $200\ \Omega$ inductive reactance.

The attenuation of the delay line terminated in a $50\ \Omega$ resistive load was measured from 50 kHz to 120 MHz. It was observed to slowly increase with frequency from zero to about 30% at 85 MHz. The total time delay T_N of the line was measured on a 7504 Tektronix oscilloscope²⁰ by applying a fast risetime step voltage from a Hewlett-Packard 213B pulse generator. A delay time of 50 nsec was obtained. The measured and computed values of T_N and Z_0 are in reasonable agreement with one another.

With this particular arrangement of the N coils making up the lumped parameter delay line, the rf field inside the delay line rapidly becomes increasingly inhomogeneous, as the wavelength of the rf wave on the delay line becomes shorter than the radius of the coil. The radial dependence of the longitudinal component of magnetic field in a helically conducting cylinder has been calculated as a function of wavelength using formulas in Ref. 21. For a magnetic field wave of form $B_z(r) \exp[i(kz - \omega t)]$ propagating along the

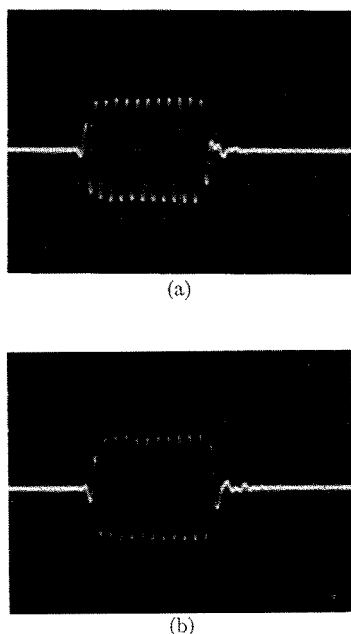


FIG. 5. (a)—rf pulse from transmitter measured at point B. The rest of the probe has been replaced by a $50\ \Omega$ termination. (b)—rf pulse from transmitter at point C after traveling through the delay line. The circuit between point C and the receiver was replaced by a $50\ \Omega$ termination. Horizontal scale— $0.1\ \mu\text{sec}/\text{div}$; vertical scale— $200\ \text{V}/\text{div}$.

direction of the cylinder, and for values of wavelength $2\pi/k$ and radius a appropriate to our delay line, the value of B_z at the boundary of the cylinder is roughly five times its value at the axis. A value of $ka=3.2$ has been assumed and end effects have been neglected. A different arrangement of coils with the radii much smaller than the length would produce a much more homogeneous rf magnetic field.

V. PERFORMANCE CHARACTERISTICS

It was mentioned in Sec. I that a resonant circuit of moderate Q rings for a time of order $\tau_r=2Q/\omega_r$ at a frequency close to the driving frequency, when excited by a rf pulse with fast rise and fall times. In comparison, a delay line with a high cut-off frequency ω_c , will have transients of short duration when excited by the same rf pulse. Moreover, these transients will no longer have frequency components near the excitation frequency, but normally at much higher ones.

Figure 5(a) shows a typical rf pulse from our transmitter. The signal is measured at point B (Fig. 4) by replacing the rest of the probe with a $50\ \Omega$ termination. Figure 5(b) shows the rf excitation pulse at point C after traveling through the delay line. For this measurement a $50\ \Omega$ termination to ground replaced the circuit between point C and the receiver. The distortion of the rf pulse by the delay line is very small allowing excellent recovery characteristics as far as transients from the delay line are concerned.



FIG. 6. Free induction decay signal using a delay line probe from protons in a polyethylene sample at 20°C . Horizontal scale— $1\ \mu\text{sec}/\text{div}$. $1\ \mu\text{sec}$ rf excitation pulse starting at the second division.

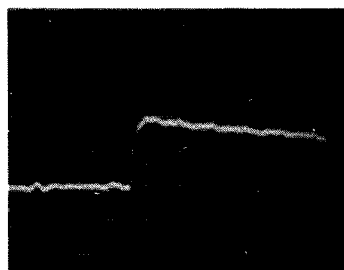


FIG. 7. Free induction decay signal from protons in polyethylene dielectric in RG-58/U cable at 20°C . Horizontal scale— $1\ \mu\text{sec}/\text{div}$. $2\ \mu\text{sec}$ rf excitation pulse starting at the first division.

Figure 6 shows the recovery characteristics and the signal-to-noise ratio attainable with the delay line probe described in Sec. IV. The signal is from the protons in a polyethylene sample at 20°C . The sample is 1 cm long and 0.6 cm diam and the Larmor frequency is 32 MHz. The total bandwidth of the receiver plus phase sensitive detector was about 4 MHz. One large horizontal graticule division in Fig. 6 corresponds to $1\ \mu\text{sec}$ in time. The excitation pulse lasts for $1\ \mu\text{sec}$ and starts $2\ \mu\text{sec}$ from the beginning of the oscilloscope trace. For a time of $1.3\ \mu\text{sec}$, starting from the beginning of the rf pulse, a Hewlett-Packard²² diode gate (model 10534A) following the receiver opens and prevents undesirable signals (the excitation pulse plus transients) from reaching the phase sensitive detector and filters. The nuclear signal can be observed without distortion less than $1\ \mu\text{sec}$ after the end of the rf pulse.

In order to determine the minimum blocking time attainable with a delay line probe used with the rest of our apparatus, the lumped parameter delay line described earlier was replaced by a piece of coaxial cable (RG-58/U) placed between the poles of the magnet. The nuclear signal observed in Fig. 7 is from the protons in the polyethylene dielectric of the coaxial cable. The recovery characteristics are now limited by transients generated in the gate that protects the receiver. The result shown in Fig. 7 indicates that the recovery time could still be improved in our delay line probe, by a more careful design of this gate.

As mentioned in Sec. IV, the oscillating magnetic field for our coil configuration is very inhomogeneous; thus there is no "true $\pi/2$ pulse width." The amplitude of our signal from the sample, when studied as a function of pulse width, increased from zero as the width increased, reached a broad maximum of $0.8\ \mu\text{sec}$, and then decreased going slightly negative and disappearing into the noise for widths greater than $3\ \mu\text{sec}$. A formula for $\bar{B}_1$, the average oscillating field inside a delay line is given by Eq. (A8). For an excitation power $P_d=900\ \text{W}$, $T_N=50\times 10^{-9}\ \text{sec}$ and $V_N=0.32\times 10^{-6}\ \text{m}^3$, one obtains a $\bar{B}_1=188\ \text{G}$. This corresponds to a rotating magnetic field of 94 G, which is capable of nutating protons

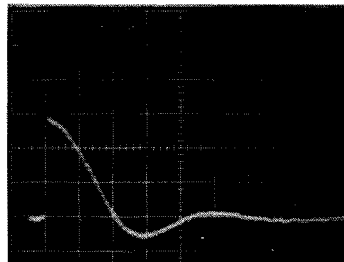


FIG. 8. Free induction decay signal from F^{19} in CaF_2 sample at $77\ \text{K}$. Magnetic field along $[100]$ crystal axis. Receiver bandwidth is 2 MHz. Horizontal scale— $10\ \mu\text{sec}/\text{div}$. $1\ \mu\text{sec}$ rf excitation pulse centered at first division.

by 90° in $0.64 \mu\text{sec}$. This result is consistent with the $0.8 \mu\text{sec}$ pulse width necessary to maximize our signal.

The fluorine signal from a single crystal of CaF_2 at 77 K is displayed in Fig. 8. Its measured peak value was $244 \mu\text{V}$ when referred to the input of the receiver. The excitation pulse had a width of $0.8 \mu\text{sec}$. A second identical pulse immediately following the first pulse yielded a signal about 30% of the initial one, indicating that not all the spins in the sample had been nutated by 90° , and that the distribution of M_x in the sample was not very homogeneous. One might expect that a uniformly excited sample might have increased the signal by about 50% and thus yielded about $340 \mu\text{V}$. The expression for the signal voltage E_N at the receiver input from a lumped parameter delay line coupled to a uniformly excited sample is given by Eq. (B8). Taking the fluorine-fluorine distance to be $2.725 \text{ \AA}$, we compute $\chi_0 = 9.7 \times 10^{-9}$ for CaF_2 at 77 K . Using Eq. (B8) with our delay line parameters and this value of χ_0 we compute $E_N = 837 k_2 \mu\text{V}$. A comparison of the computed and experimental values of E_N would set the value of $k_2 \approx \frac{1}{2}$, which is reasonable, considering the geometry involved.

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APPENDIX A: COMPUTATION OF OSCILLATING MAGNETIC FIELD IN THE DELAY LINE

The peak magnetic energy $\mathcal{E}_1$ stored in inductance L_1 in Fig. 1 is

$$\mathcal{E}_1 = \frac{1}{2} (I_n') (I_n')^* L_1, \quad (\text{A1})$$

$$\mathcal{E}_1 = \frac{1}{2\mu_0} \iiint B_1^2(\mathbf{r}) dv, \quad (\text{A2})$$

where $B_1(\mathbf{r})$ is the peak value of the *oscillating* magnetic field at position $\mathbf{r}$ due to current I_n' passing through L_1 , and $\mu_0 = 1.257 \times 10^{-6} \text{ W/A-M}$. From Eq. (A1) and Eq. (40), the power P_D traveling down the transmission line is computed to be

$$P_D = \mathcal{E}_1 \left(\frac{\omega}{2} \right) \left(\frac{\sin b}{1 - \cos b} \right) \quad (\text{A3})$$

which for $\omega \ll \omega_c$ reduces to

$$\mathcal{E}_1 = P_D T_1. \quad (\text{A4})$$

Assume that coil L_1 has a volume V_1 inside of which the average magnetic field is $\bar{B}_1$. Then

$$\mathcal{E}_1 = \frac{1}{f_1^2} \frac{1}{2\mu_0} V_1 \bar{B}_1^2, \quad (\text{A5})$$

where

$$f_1^2 = (V \bar{B}_1^2) \left[\iiint B_1^2(\mathbf{r}) dv \right]^{-1}, \quad (\text{A6})$$

f_1 is dependent only upon the geometry of the coil, is always

less than 1, and is typically assumed to be $\frac{1}{2}$ for many experiments. Combining (A4) and (A5) yields

$$\bar{B}_1 = f_1 \left(\frac{2T_1 P_D \mu_0}{V_1} \right)^{\frac{1}{2}} \quad (\text{A7})$$

$$\bar{B}_1 = f_1 \left(\frac{2T_N P_D \mu_0}{V_N} \right)^{\frac{1}{2}}, \quad (\text{A8})$$

where $V_N = NV_1$.

APPENDIX B: COMPUTATION OF THE SIGNAL VOLTAGE DELIVERED BY A DELAY LINE TO THE RECEIVER

The signal power P_D delivered by a lumped parameter delay line terminated in its characteristic impedance to a receiver is given by Eq. (49). For $\omega \ll \omega_c$, it reduces to

$$P_D = \frac{N^2 \epsilon_0^2 T_1}{8L_1}. \quad (\text{B1})$$

It will now be assumed that each section of the delay line has a cross sectional area A_1 , a length ℓ_1 , a volume $V_1 = A_1 \ell_1$, n_1 turns and a sample of material with a uniformly distributed magnetization/unit volume of M_x that oscillates at the frequency ω . Ignoring leakage flux, one gets the relationship

$$\frac{n_1}{\ell_1} = \left(\frac{L_1}{V_1 \mu_0} \right)^{\frac{1}{2}}. \quad (\text{B2})$$

Assuming that the flux from the magnetization M_x couples all n_1 turns yields

$$\phi = \mu_0 M_x A_1, \quad (\text{B3})$$

$$\begin{aligned} \epsilon_0 &= n_1 \left| \frac{d\phi}{dt} \right| \\ &= \mu_0 n_1 A_1 M_x \omega \\ &= M_x \omega (L_1 \mu_0 V_1)^{\frac{1}{2}}. \end{aligned} \quad (\text{B4})$$

To account for the decrease in signal due to the ignoring of leakage flux in computing Eq. (B2) and the flux from the sample not coupling all n turns, one introduces a factor k_2 into Eq. (B4) so that it becomes

$$\epsilon_0 = k_2 M_x \omega (L_1 \mu_0 V_1)^{\frac{1}{2}}. \quad (\text{B5})$$

k_2 is dependent only upon the geometry of the coil; it is always less than one and is typically taken to be $\frac{1}{2}$ for many experiments. Substituting Eq. (B5) into (B1) yields

$$P_D = \frac{\mu_0 M_x^2 \omega^2 V_N T_N k_2^2}{8}. \quad (\text{B6})$$

If one assumes that $M_x = (1/\mu_0) \chi_0 B_0$, where χ_0 is the nuclear susceptibility of the sample, and $\omega = \gamma B_0$, where B_0 is the applied static magnetic field and γ is the mag-

netogyric ratio of the nuclei then,

$$P_D = \frac{\chi_0^2 \omega^4 V_N T_N k_2^2}{8\gamma^2 \mu_0} \quad (B7)$$

and

$$E_N = (2Z_0 P_D)^{\frac{1}{2}} = \frac{\chi_0 \omega^2 k_2}{2\gamma} \left(\frac{Z_0 V_N T_N}{\mu_0} \right)^{\frac{1}{2}} \quad (B8)$$

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